

PetInsightPro

Group Member:

Jingwen Ni Wanyu Lin

Yucheng Song Ruze Shen Bowen Lu



Project Overview

Project Name: PetInsightPro

PetInsightPro is China's **first** smart wearable device focused on "**early pet health screening + veterinary diagnostic support**," **redefining** pet health management.

▪ **Current stage**

-Completed **technical feasibility** research; **software development** is underway and expected to finish in **1-2 months**.

▪ **Market opportunity**

-China has **126 million** urban pet dogs and cats, and **87%** of owners are interested in smart devices, but existing products **lack** professional tools.

▪ **Business model**

-Combines **low-cost** hardware, **subscription** services, and hospital referral fees.

▪ **Collaboration**

- seeks **medical records** and clinical annotations from veterinary **hospitals** in **exchange** for free **devices**, **referral fees**, and **joint authorship**.

Problem

Whether fur babies are healthy or not?

- The average cost is **2,786 RMB**

- For **90%** of diseases, early intervention results in significantly lower treatment costs
- **71.7%** of trauma-related deaths in cats occur within the first 24 hours



Survey

问卷标题:宠物智能项圈市场调研



长按识别二维码

Sample Size	200 valid responses
Owner Age	20–30 years (57%), 30–40 years (36%)
Pet Age	0–1 year (57%), 1–5 years (29%), 10+ years (21%)
Annual Health/Safety Spending	1,500–3,000 RMB (43%), 3,000–5,000 RMB (37%) 500–1,500 RMB (20%)
Top Pet Concerns	1. Sudden illness undetected (71%) 2. Lost or stolen (64%) 3. Unsafe diet (64%)
Willingness to Pay for Health Alerts	Very willing (53%) + Somewhat willing (43%) = 96%
Most Preferred Product Version	Pro Max (43%) > Pro (29%) > Normal (21%)
Reasonable Price Range	• Normal: 200–350 RMB (79%) • Pro: 350–500 RMB (64%) • Pro Max: 500–700 RMB (86%)
Social Feature ("Pet Social Circle")	86% find it interesting (43% very + 43% somewhat) → good differentiator
Main Purchase Triggers	1. Pet previously lost or sick (64%) 2. Friends using it (43%) 3. Ads or reviews (36%)
Final Purchase Intent	87% would buy

- Focus on **Pro Max (500–700 RMB)**
- Emphasize **health alerts + anti-loss**
- **Social** features appeal only to dog owners.; all pet owners want **health monitoring**

Market



The number of pet dogs and cats in reached **126 million**



64% pet owners are
Interested in smart devices



Pet healthcare market size:
growing at over **15%** annually



Birth rate **5.63‰**
People prefer pets to children

Target Users

Individual Users

Urban white-collar workers aged 20 to 45, busy lifestyle, first-time pet owners or those with experiences of pet loss or illness

Pain 1: Hidden diseases hard to detect early

Pain 2: Round-the-clock monitoring unavailable after surgery or during recovery

Pain 3: Subjective descriptions available instead of objective medical data

Pet Hospitals

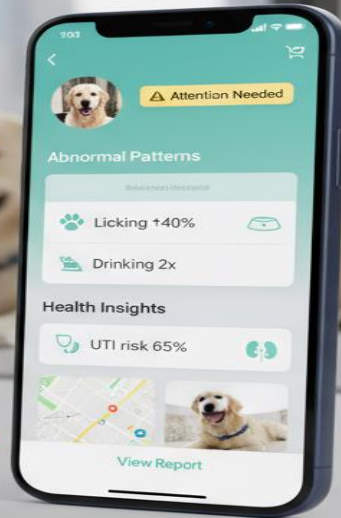
Pain 1: Post-operation and chronic disease management rely on owners' subjective accounts

Pain 2: Lack of effective contact with owners

Pain 3: Insufficient continuous clinical behavioral data restricts medical research and treatment improvement

Solution

**GPS
Collar, Monitor
Camera:**



PetInsightPro — Know before it's serious

Normal:

GPS positioning + basic health monitoring

Pro:

Normal + abnormal behavior analysis

Pro Max:

Pro + HD camera (records surroundings of the pet)

Solution

Three-Tier Architecture

Hardware Sensing Layer:

- capture multimodal physiological data via low-power optical heart rate sensors (PPG), infrared thermometry, and a 6-axis IMU

AI Algorithm Layer:

- leveraging Edge AI (TinyML), the collar processes raw data locally to classify 19+ behavioral patterns with minimal latency
- the cloud backend runs risk-assessment models

Data Service Layer:

- use Python for automated data cleaning for raw telemetry
- deploy Tableau-driven Business Intelligence dashboards to monitor user engagement metrics and generate medical-grade health reports

Current Product Landscape

Type	Representative Products	Core Functions	Price Range
GPS Anti-loss	Peipei, Xiaopei Pendant, Xiaomi	Real-time positioning, electronic fence	100-350 yuan
Health Monitoring	Whistle, Tractive, Xiaopei Partner	Heart rate / body temperature / activity monitoring	500-1300 yuan + subscription
AI Emotional Collar	PurrPurr, etc.	AI emotion recognition, behavior analysis	499 yuan + subscription

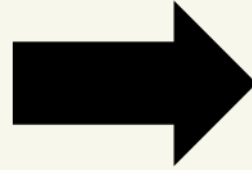
Market Gap: There are currently no products focusing on **"Veterinary-assisted Diagnosis + Health Reports"**.

Business Model ---how do we make money

Products



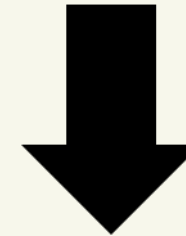
Hardware
Acquisition



Customers



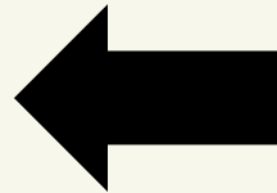
Subscription
Revenue



Great Feedback



App
Engagement



Community



Data-Driven
Improvement



Competitive Landscape and Core Barriers

Differentiated Advantages



Precise Medical Assistance Positioning

Different from ordinary consumer electronics, it clearly focuses on "veterinary-assisted diagnosis", providing professional-level data support and establishing professional barriers in the niche field.



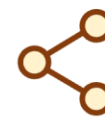
"Health Report" as a Core Necessity

Directly addresses pet owners' pain points by solving the problem of being unable to provide **objective** and **detailed data** during medical treatment. The professional health report can be **directly** used as an important reference.



Highly Attractive Business Model

Adopts a "**low-cost hardware + subscription service**" combination: 298 CNY affordable price lowers the threshold, and 29.9 CNY/month service creates **stable recurring revenue**.



Efficient User Growth Mechanism

Uses a "**7-day free trial**" to reduce decision costs, and acquires users quickly at **low cost** through **social fission gameplay**, building a **growth flywheel**.

Cooperation Model with Pet Hospitals

Core Content and Value Proposition for Cooperation

01 / Cooperation Content



Data Collection Cooperation

Invite doctors to record **real** diagnosis, or **label** the **correspondence** between specific behaviors and symptoms to improve the model.



Channel Promotion Cooperation

Recommend products to customers, hospitals get **15-20%** one-time **promotion fee**.



Consulting Fee Sharing (Long-term)

Long-term share of **20-30%** from online **consultation fees**.



02 / Multi-dimensional Value for Hospitals



Direct Economic Benefits

Easy to get promotion fees and consulting shares.



Research Capability

Free equipment & reports, support co-authorship.



Diagnostic Quality

Health reports assist in accurate diagnosis.



Reduction of Risks

Remote monitoring to detect abnormalities.

Revenue Engine

DTC Social Marketing

- To minimize customer acquisition cost, initial marketing will heavily leverage high-contrast visual plogs and short-form video content on platforms such as Rednotes and Instagram.
- This capitalizes on the aesthetic preferences of Gen-Z pet owners while addressing their deep-seated anxiety regarding hidden pet illnesses.



Customer Acquisition (35%-45% Margin)

Hardware is competitively priced to lower the barrier to entry, serving primarily as a data-gathering terminal & high-conversion hook.

LTV Maximization (70%-80% Margin)

The core profit engine relies on recurring monthly/annual subscriptions (¥29.9/mo), offering premium features.

Go-to-market —how can we expand the market



User Referral & Growth



Partner with veterinary clinics



Promote on social media



Long-term Retention



Roadmap

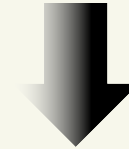
Step 0 - Current Market Research & Validation

- in-depth interviews
- market research
- competitive analysis
- literature review



Step 1 - Launch Software (Frontend + Backend) & Basic Collar Design

- **Backend:**
 - health **data storage**
 - **initial algorithms** for behavior
 - **Mobile App:**
 - Real-time GPS **tracking** ...
 - **Basic collar design:**
 - GPS + 6-axis gyroscope + heart rate sensor ...
- internal testable version



Step 2 - Grow Feature Completion

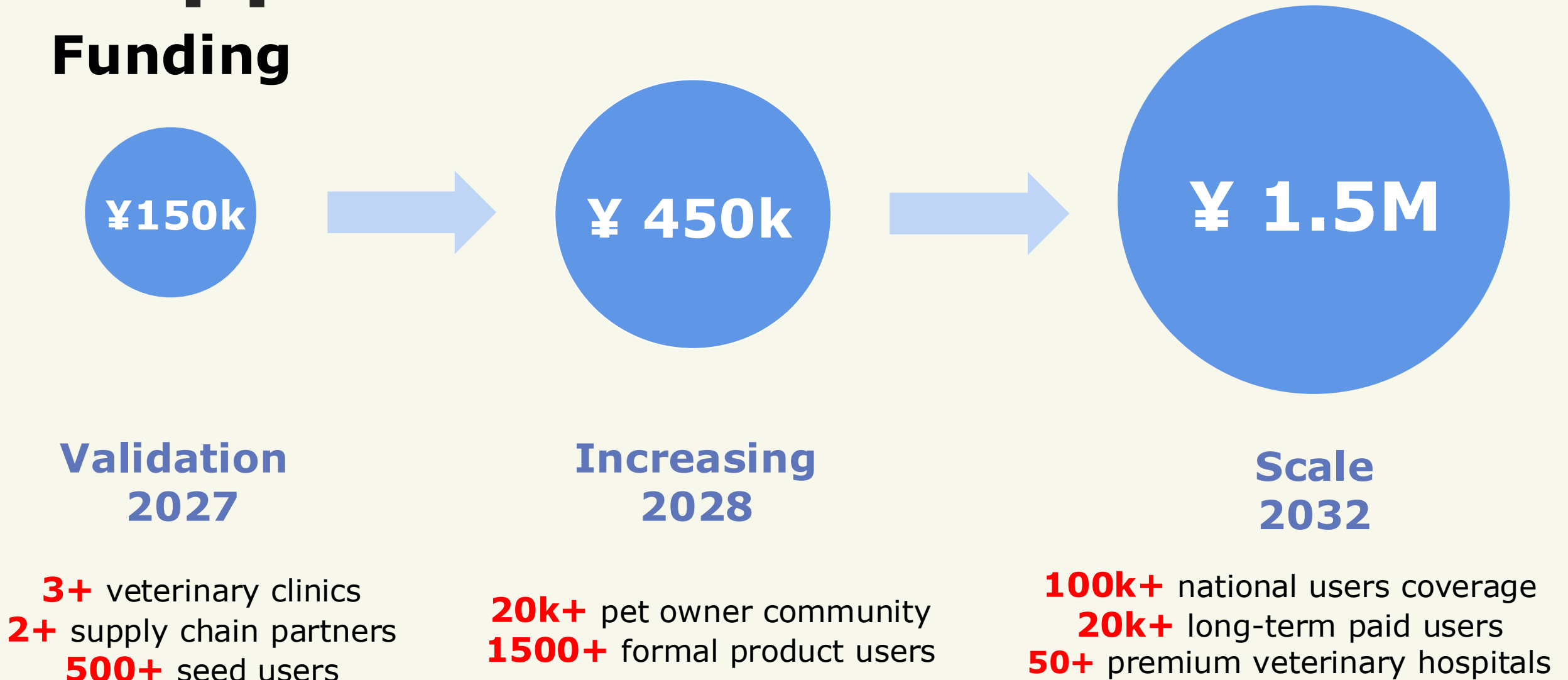
- **Add key functions:**
 - Tamper-proof **alarm** ...
 - **Improve algorithm accuracy:**
 - Abnormal behavior **detection**
 - Heart rate anomaly triggers
- small batch engineering prototypes + full-feature app



Step 3 - Scale User Data Feedback → Product Iteration

- **Collect feedback on:**
 - **usability** of health reports
 - app **experience**
 - **Iteration focus:**
 - Reduce false alarms
- market-ready stable version

Support Funding



Support Incubation

Basic Office:

Address --- establish a company

Room --- store equipments

Sponsors:

30k RMB certification and legalization

Industrial Resource:

OEM factories for PetSmart hardware
core **component** suppliers

